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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CV2005-1 – DESIGN OF RC STRUCTURAL ELEMENTS

(For CIV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Use of IS:456-2000 is permitted.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Reinforced concrete (RC) is primarily advantageous because:
A) It is cheap
B) Concrete is strong in tension
C) Concrete is strong in compression and steel is strong in tension
D) It does not require curing
2. IS 456:2000 deals with:
A) Structural steel design
B) Design of Reinforced Concrete Structures
C) Masonry design
D) Prestressed concrete design
3. The Working Stress Method (WSM) is based on:
A) Limiting stresses in materials under working loads
B) Limit state of collapse
C) Ultimate loads
D) Plastic analysis
4. In Limit State Method (LSM), design is based on:
A) Working loads only
B) Ultimate loads and characteristic material strengths
C) Experimental data only
D) Stress-strain graphs only
5. The ultimate flexural strength of a singly reinforced RC section depends on:
A) Area of steel and depth of neutral axis
B) Concrete tensile strength
C) Shear strength
D) Creep of concrete
6. Doubly reinforced sections are provided when:
A) Concrete alone resists the load
B) To reduce deflection
C) Compression reinforcement is needed to increase ultimate moment capacity
D) For aesthetic purposes
7. The stress block parameters in IS 456:2000 for limit state of collapse include:
A) Modulus of elasticity and Poisson's ratio
B) Only characteristic load
C) Factors for stress distribution
D) Ultimate shear stress
8. Development length in RC is required to:
A) Ensure adequate anchorage of steel to develop full strength
B) Reduce beam width
C) Reduce concrete cover
D) Increase column slenderness
9. Serviceability Limit State checks for:
A) Ultimate load
B) Shear failure only
C) Concrete strength only
D) Deflection, cracking, and durability



10. Deflection limits in RC members are specified in:
 A) IS 800 B) IS 456:2000
 C) IS 875 D) SP 16
11. Cracking in structural concrete members is controlled by:
 A) Limiting tensile stress in steel and spacing of bars B) Using plain concrete
 C) Increasing beam depth only D) Reducing concrete cover
12. In beam design, the critical section for shear is located at:
 A) At one-fourth span B) Midspan
 C) At the centroid D) Supports
13. The effective length of a column is used to determine:
 A) Shear strength B) Slenderness ratio and buckling risk
 C) Flexural strength D) Deflection
14. Minimum eccentricity of axial load in columns is considered to:
 A) Prevent accidental concentric loading and buckling B) Reduce bending moment
 C) Increase column height D) Reduce cross-section area
15. SP 16 provides guidelines for:
 A) Design of steel structures B) Design aids for Reinforced Concrete as per IS 456:2000
 C) Masonry construction D) Prestressed concrete design
16. A short axially loaded column is one in which:
 A) Slenderness ratio < 12 B) Slenderness ratio > 12
 C) Slenderness ratio = 15 D) No slenderness check is required
17. A one-way slab primarily bends in:
 A) Both directions equally B) Along the longer span
 C) Only at supports D) Along the shorter span
18. The design of a cantilever slab requires:
 A) Moment at midspan only B) Shear only at midspan
 C) Negative moment at the fixed support D) No reinforcement
19. For a two-way slab, bending occurs:
 A) In both directions B) In one direction only
 C) Only at supports D) Only at midspan
20. The practical requirement in slab design includes:
 A) Maximum concrete volume only B) Ignoring deflection
 C) Minimum reinforcement for crack control D) Using steel without concrete

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the Working Stress Method (WSM) and Limit State Method (LSM) of design as per IS 456:2000. Highlight their differences, advantages, and limitations	8	L*2	1	1
b)	A singly reinforced beam has width 300 mm and effective depth 500 mm, reinforced with 3 bars of 20 mm diameter Fe500 steel. Calculate the ultimate moment of the section using the Limit State Method.	8	L2	2	2
2. a)	Explain the concept and need for doubly reinforced beams.	6	L2	1	1
b)	A doubly reinforced rectangular beam has width 250 mm, effective depth 450 mm, compression reinforcement 2 bars of 16 mm diameter, tension reinforcement 3 bars of 20 mm diameter. Calculate the ultimate moment capacity.	10	L2	2	2

3. a) Explain the ultimate shear strength of RC beams and the IS 456:2000 procedure for shear design.
- b) A beam 300 mm × 500 mm carries a factored shear of 150 kN. Determine the required shear reinforcement if the concrete contribution is 60 kN.

6	L2	2	1
10	L2	3	2

Unit – II

4. a) Explain the practical design requirements and critical sections for bending and shear in rectangular beams.
- b) Design a simply supported RC beam of width 300 mm × depth 500 mm subjected to a factored bending moment of 250 kNm.
5. a) Discuss general aspects of column design, effective length, slenderness ratio, and minimum eccentricity.
- b) Design a short square column 400 mm × 400 mm carrying an axial factored load of 1200 kN.
6. a) Explain the design procedure for columns subjected to combined axial load with uniaxial and biaxial bending.
- b) Design a rectangular column of size 400 mm × 600 mm, subjected to a factored axial load of 1500 kN, factored bending moments of 80 kNm about the x-axis and 60 kNm about the y-axis. Use $f_{ck} = 30$ MPa and $f_y = 500$ MPa.

6	L2	3	1
10	L2	3	3
6	L2	3	1
10	L2	4	3
6	L2	4	1
10	L2	4	3

Unit – III

7. a) Discuss practical requirements and design considerations for one-way, two-way, and cantilever slabs.
- b) Design a simply supported one-way slab of span 4 m, 1 m width, carrying a dead load of 2 kN/m² and live load of 3 kN/m². Show the reinforcement layout.
8. a) Explain the design procedure and detailing of cantilever and two-way slabs as per IS 456:2000.
- b) Design a cantilever slab 1 m wide and 1.5 m long, subjected to a uniformly distributed load of 4 kN/m².

6	L1	5	1
10	L2	5	3
6	L2	5	1
10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
